Data Science for Climate Action: Predicting Agricultural Emissions for Global Regions Using ML

Agenda

The Challenge: Climate Change & Agriculture

Our Approach: Data Science & The Medallion Architecture

Data Deep Dive: Exploratory Data Analysis (EDA)

Modeling Strategy: From Classical to Modern Time Series Forecasting

Interactive Tools: Dashboards for Actionable Insights

Conclusion & Impact

The escalating climate crisis demands innovative action across all sectors.

Agriculture is the bedrock of global food security but also a major contributor to greenhouse gas (GHG) emissions.

Key activities like livestock digestion, manure management, and fertilizer use release potent GHGs like Methane (CH4) and Nitrous Oxide (N2O).

The Global Dilemma: How can we feed a growing world population without compromising the health of our planet?

Project Objectives

- To harness the predictive power of machine learning to forecast agricultural GHG emissions for distinct global regions.
- To build and rigorously evaluate a suite of time-series models (such as ARIMA, Prophet, LSTM, etc.) for forecasting agricultural greenhouse gas (GHG) emissions. The primary goal is to compare their predictive performance on data from distinct global regions to identify the most accurate and reliable methodologies to move from a reactive to a strategic posture in climate policy.
- To develop and deploy a set of interactive dashboards that visualize historical GHG emission trends and future forecasts generated by the models. This tool is designed to translate complex data into clear, actionable insights for stakeholders.

Project Architecture: The Medallion Lakehouse

Bronze Layer: Contains raw, unaltered data directly from the source.

Silver Layer: Contains validated, cleaned, and standardized data from the Bronze layer.

Gold Layer: Refined data, ready for analytics and modeling.

Bronze Layer: Raw Data Ingestion

Purpose: Contains raw, unaltered data directly from the source.

Format: Data is loaded from source CSVs and stored in an efficient Parquet format.

Process: This layer serves as a historical archive of all incoming data.

Silver Layer: Cleaned & Transformed Data

Purpose: Contains validated, cleaned, and standardized data from the Bronze layer.

Process:

- Relevant columns are selected.
- Column names are standardized.
- Multiple emission dataframes are integrated.
- Units are standardized to tonnes.
- Missing values and inconsistencies are handled.

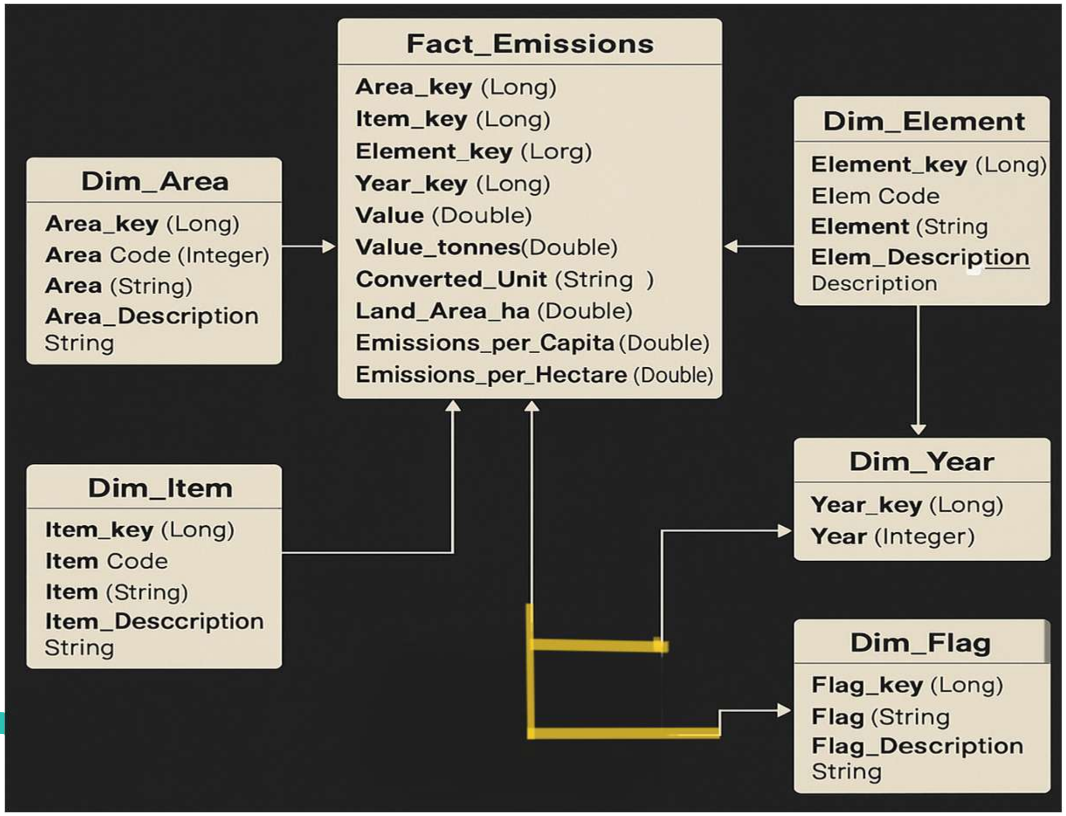
Gold Layer: Business-Ready Aggregates

Purpose: Refined data, ready for analytics and modeling.

Structure: Organized into a star schema with a central Fact Table (key metrics) and multiple Dimension Tables (attributes like Country, Year).

Benefit: This schema is optimized for high-performance queries, reporting, and ML feature generation.

Data Schema: A Star Schema Design

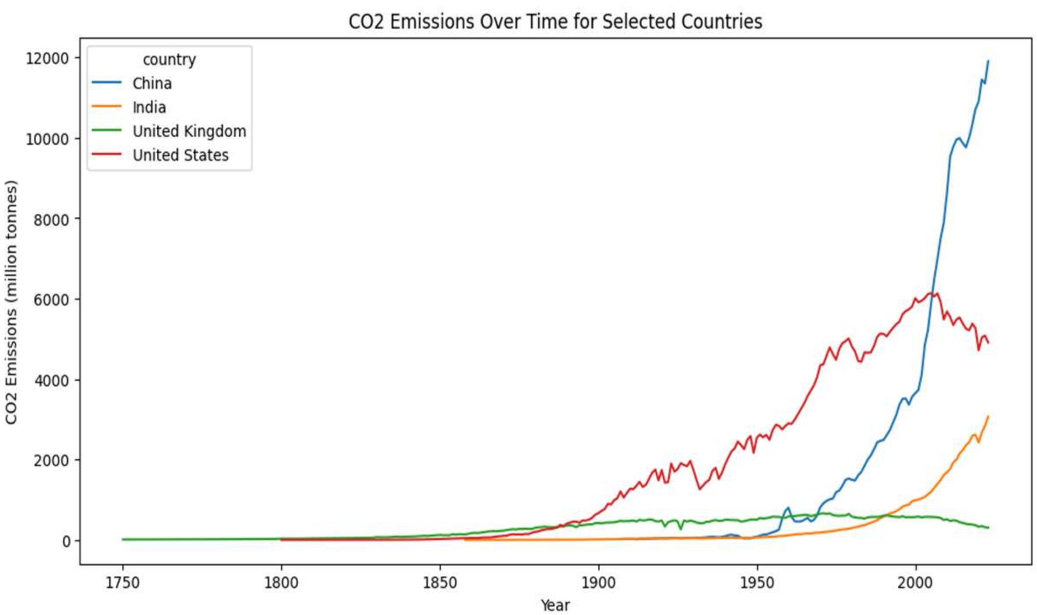


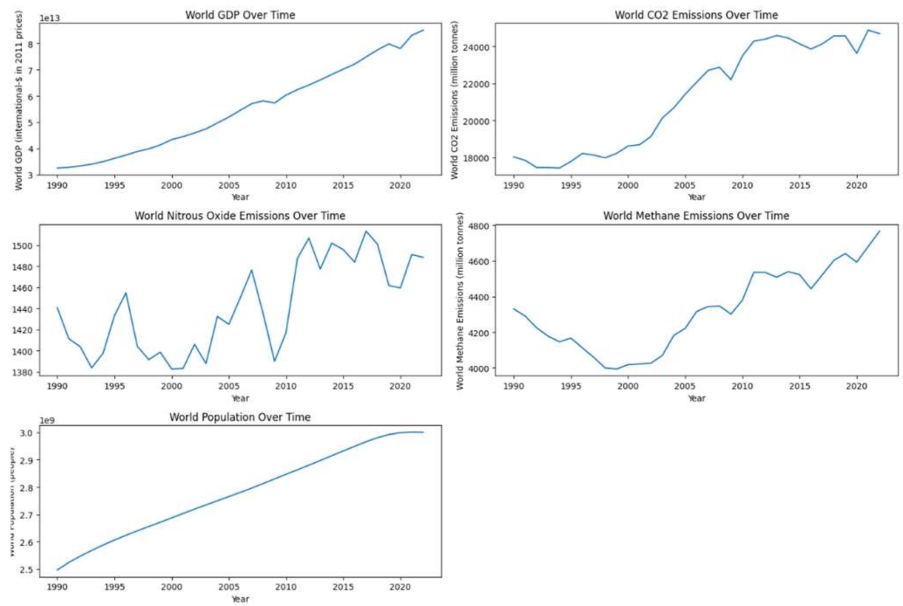
Gold layer is structured as a star schema, which clearly defines relationships and improves query performance.

Technology Stack & Data Source

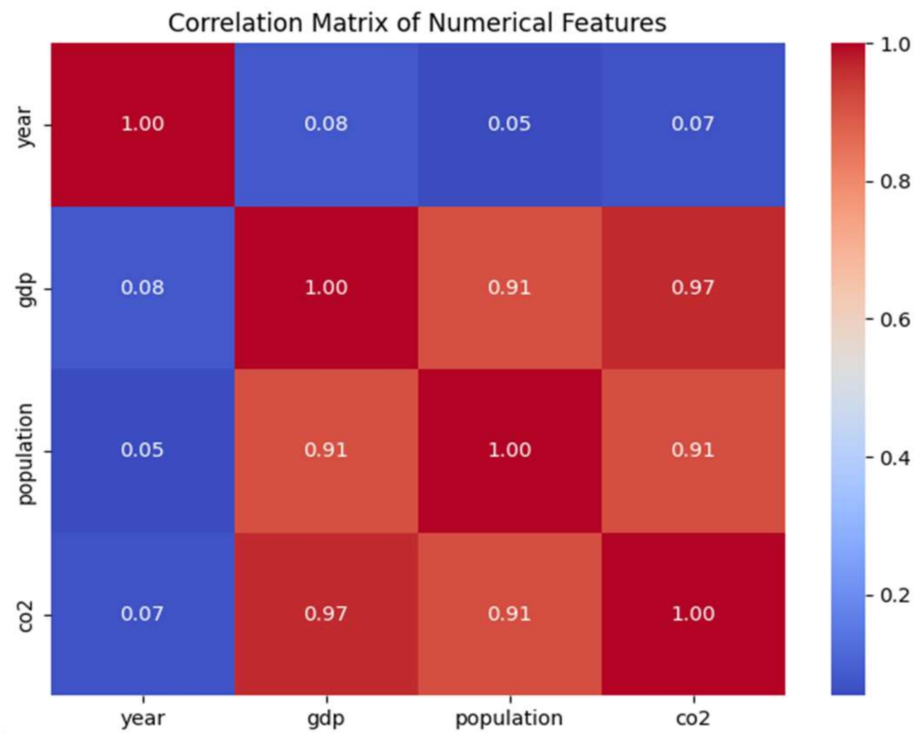
- **Programming Language:** Python 3.10, PySpark
- **IDE/Notebooks:** Google Colab, Jupyter
- **Cloud Platform:** Google Drive, Auto-scaling clusters
- **ML & Analytics Stack:** Scikit-learn, LSTM, Pandas, NumPy , Prophet
- **Deployment:** Netlify
- **Visualization:** Python/Plotly Dash
- **Data Sources: Our World in Data:** Provides country-level data on population, GDP, and total greenhouse gas emissions https://raw.githubusercontent.com/owid/co2-data/master/owid-co2-data.csv

Initial Data analysis





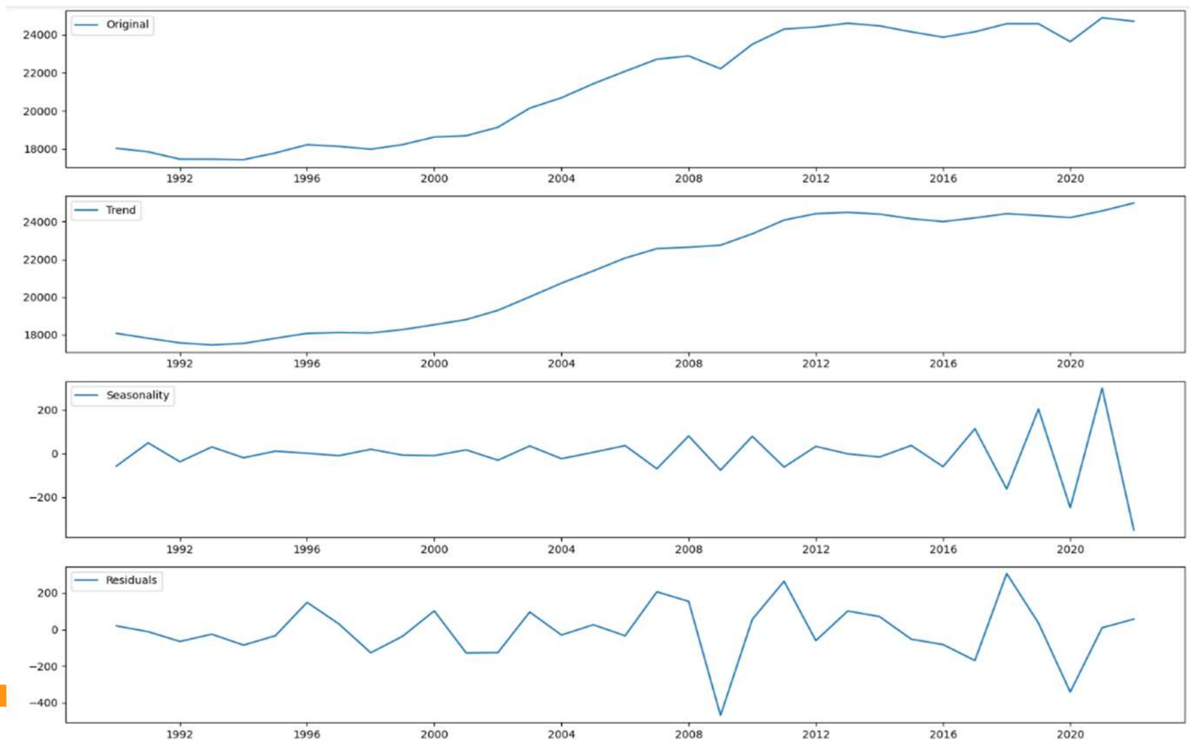
EDA: High Inter-Dataset Correlations



A correlation matrix reveals strong positive relationships between GDP, population, and CO2 emissions.

This confirms that economic and population growth are fundamental drivers of emissions

Pre-Modeling: Time Series Decomposition



Trend: The long-term direction of the data.

Seasonality: Predictable, repeating patterns.

Residual: Random, irregular fluctuations (noise).

Types of Decomposition test –

1) Classical Time series

2) STL (Seasonal and Trend decomposition using LOESS) decomposition

Pre-Modeling: Stationarity Testing

- *Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test*: used to complement the ADF test because it has the opposite hypotheses.

- *Null Hypothesis (H0):* The time series is stationary around a deterministic trend.
- Alternative Hypothesis (H1): The time series is non-stationary (it has a unit root).
- Interpretation: If the p-value is less than the significance level, reject the null hypothesis and conclude the series is non-stationary.

Augmented Dickey-Fuller (ADF) Test: This is one of the most common tests.

Null Hypothesis (H0): The time series is non-stationary.

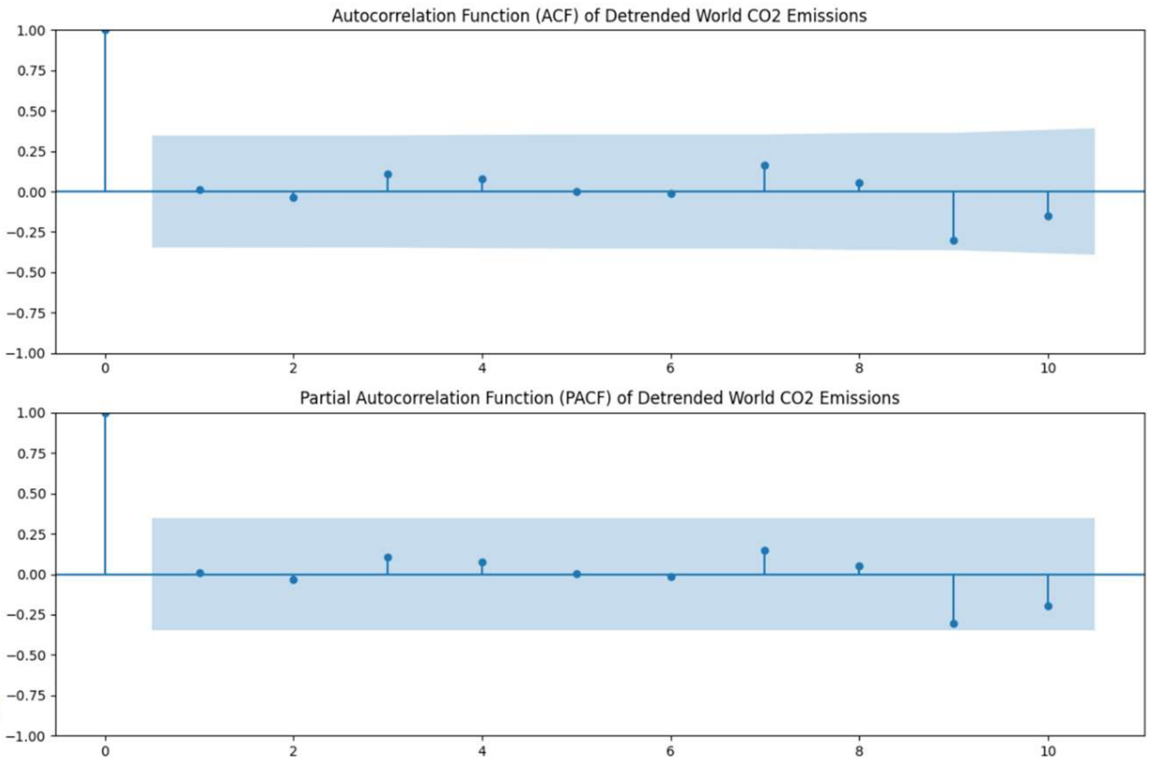
Alternative Hypothesis (H1): The time series is stationary.

Interpretation: If the p-value is less than a chosen significance level (e.g., 0.05), reject the null hypothesis and conclude the series is stationary.

Example - Original Series: Non-Stationary (p-value > 0.05)

After Detrending (1st order differencing): Stationary (p-value < 0.05)

Pre-Modeling: Autocorrelation Analysis



ACF and PACF plots of the detrended series show very few significant lags.

Inference: After first-order differencing, the series is very close to white noise. This suggests that past values have very little predictive power for future values, posing a challenge for simple models.

Modeling Approach 1: General ML Regressors

- Transformed the time series problem into a regression problem by engineering features like:
- Value_lag1, Value_lag2, Value_lag3 (past values)
- Rolling_Mean_3Y (smoothed trend)
- Compared three models: Linear Regression, Random Forest, and Gradient-Boosted Trees.

Linear Regression: Performed poorly (R^2: -1.245), unable to capture non-linearities.

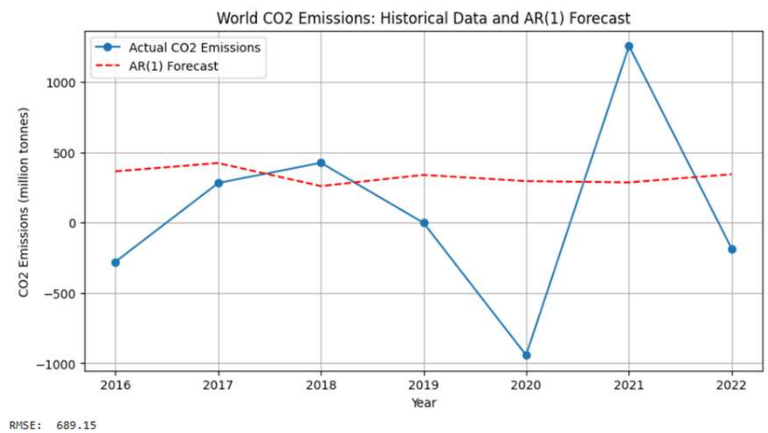
Random Forest: Performed much better (R^2: 0.040), with a lower RMSE and more realistic non-negative forecasts.

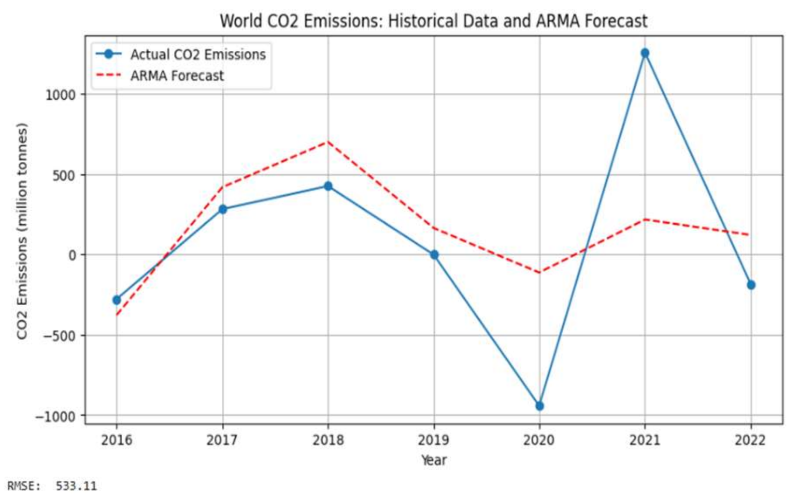
GBT Regressor: Performed poorly (R^2: -1.792), indicating overfitting.

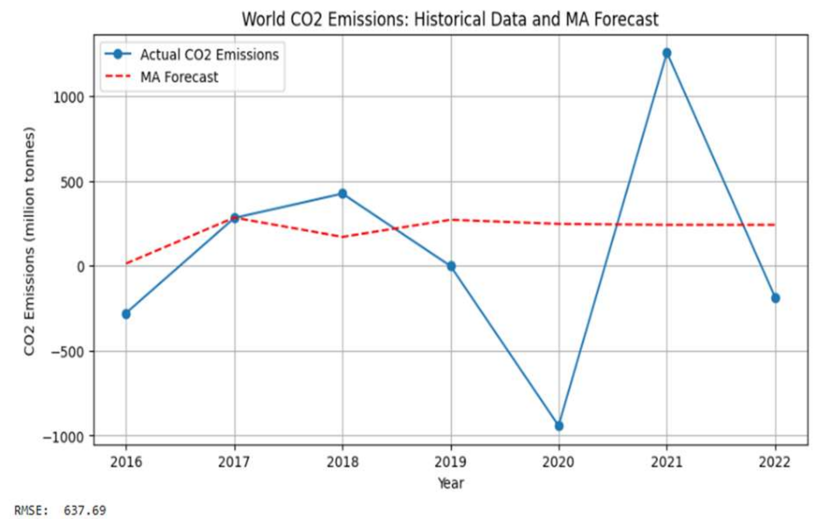
Conclusion: Random Forest was the most suitable general-purpose regressor for this task.

Modeling Approach 2: Univariate Time Series Models

- These models use only the past values of the time series itself for forecasting.
- AR (Autoregression): Uses past values.
- MA (Moving Average): Uses past forecast errors.
- ARMA: Combines both AR and MA.
- ARIMA: Extends ARMA to handle non-stationary data.







Results: ARIMA Model

- Tested several ARIMA models (e.g., ARIMA(1,1,0), ARIMA(0,1,1), ARIMA(1,1,1)).
- Inference: All ARIMA models produced a flat forecast.
- Why? Because the differenced series (the "I" part) was close to white noise, the model found no predictable pattern in the year-over-year changes. The best prediction for the future was simply the last known value.

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Visualizing ARIMA(1, 1, 1) Forecasts...
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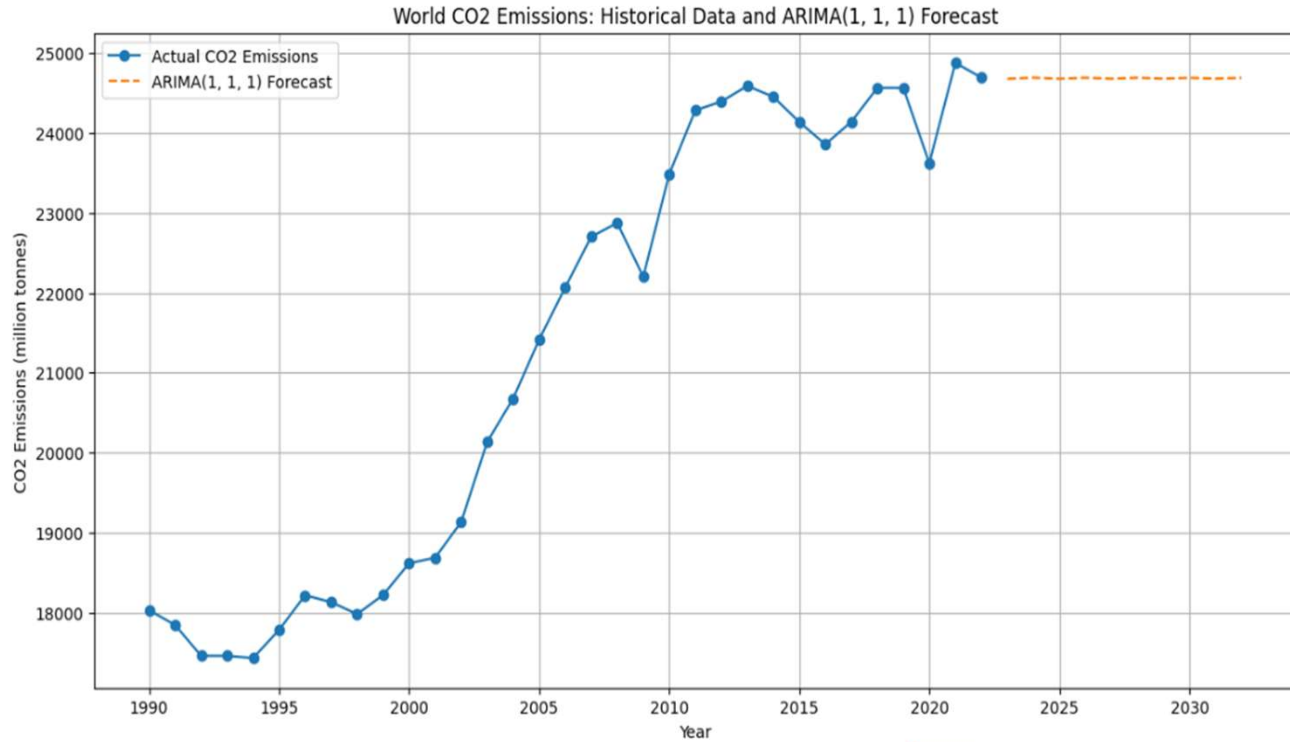
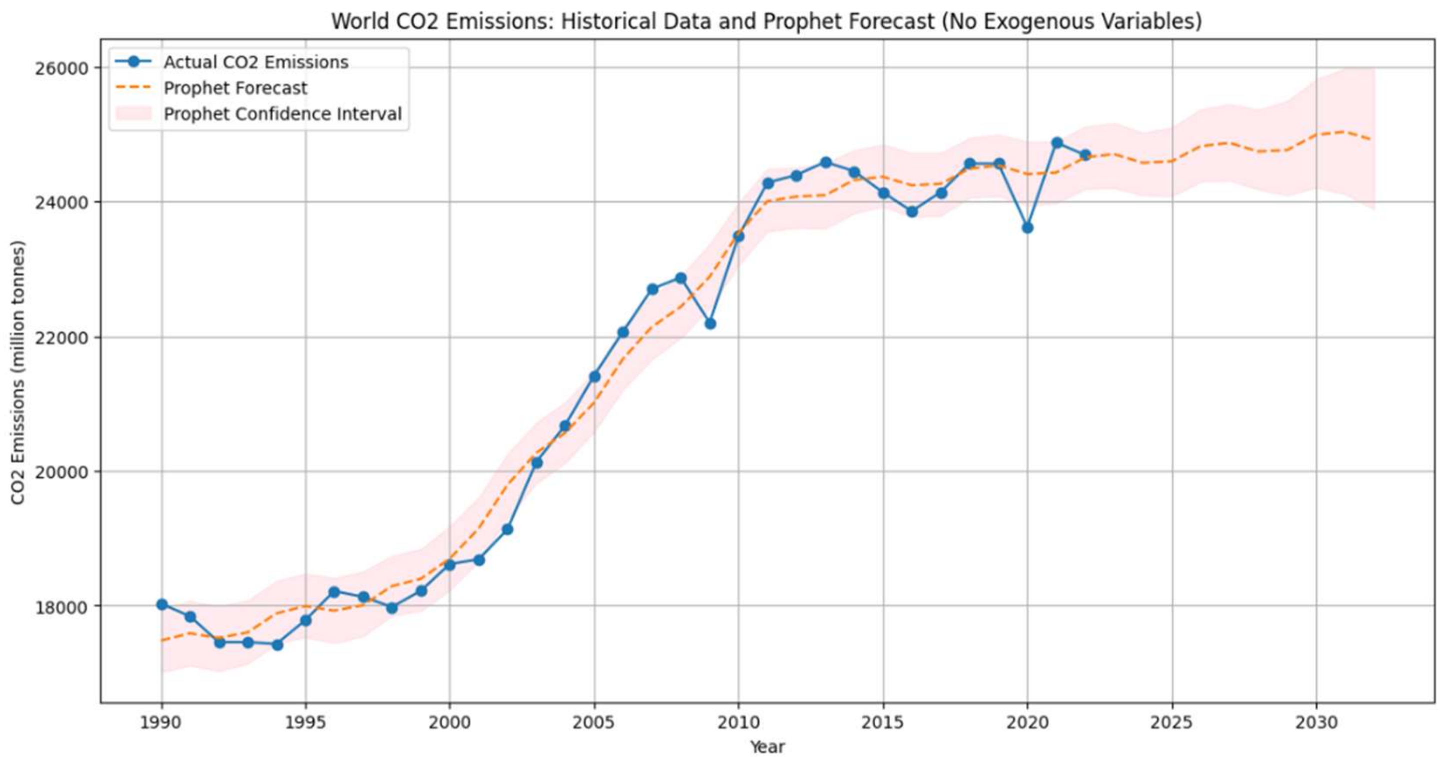


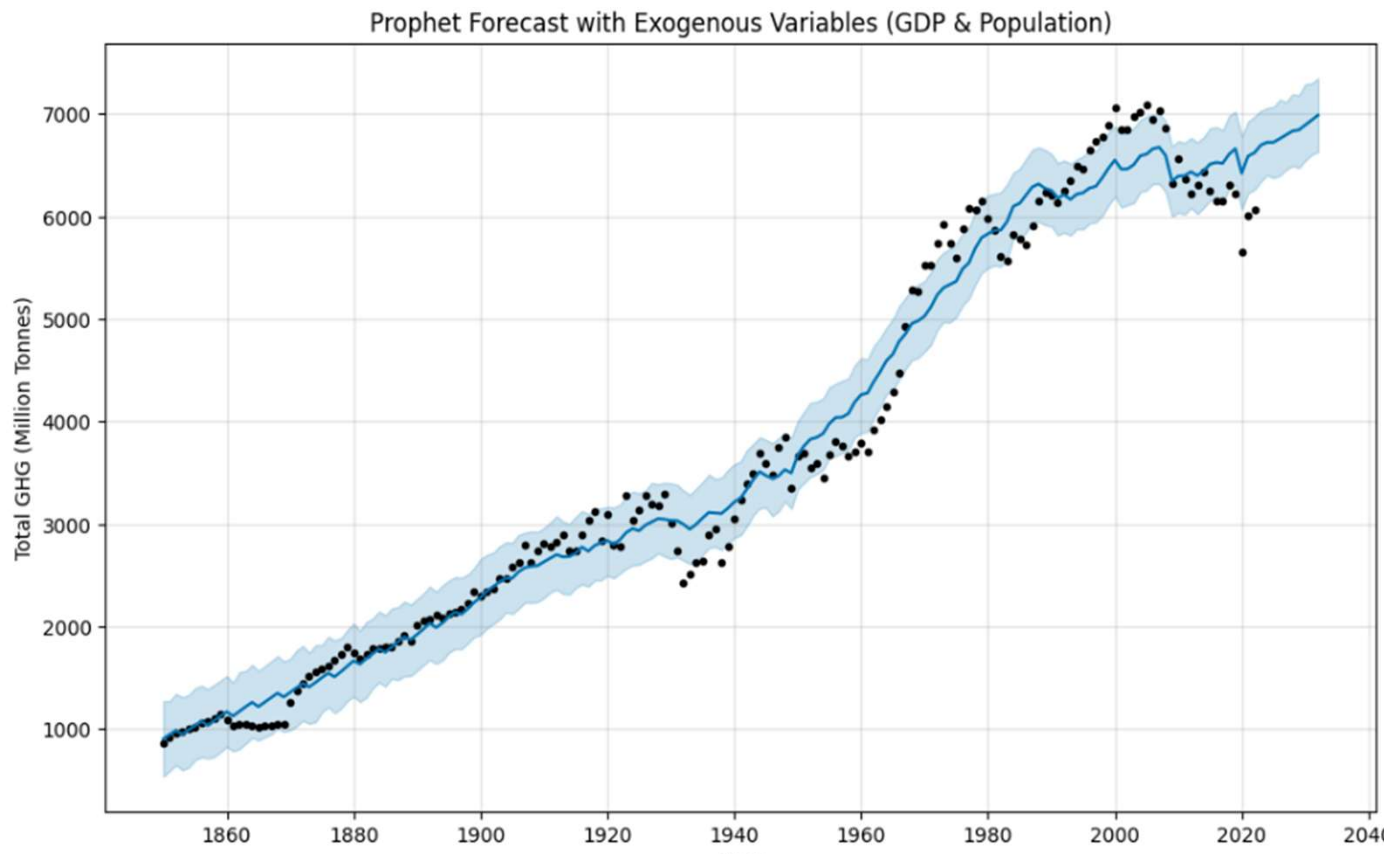
Modeling Approach 3: Prophet (without Exogenous Variables)

- Prophet is a modern, decomposable time series model develope by Facebook.
- Inference: Without external factors, Proph forecasted a slight plateau with minimal growth.
- It performed better thа ARIMA by capturing sc underlying trend, but i did not predict a continuation of the historical steep increas



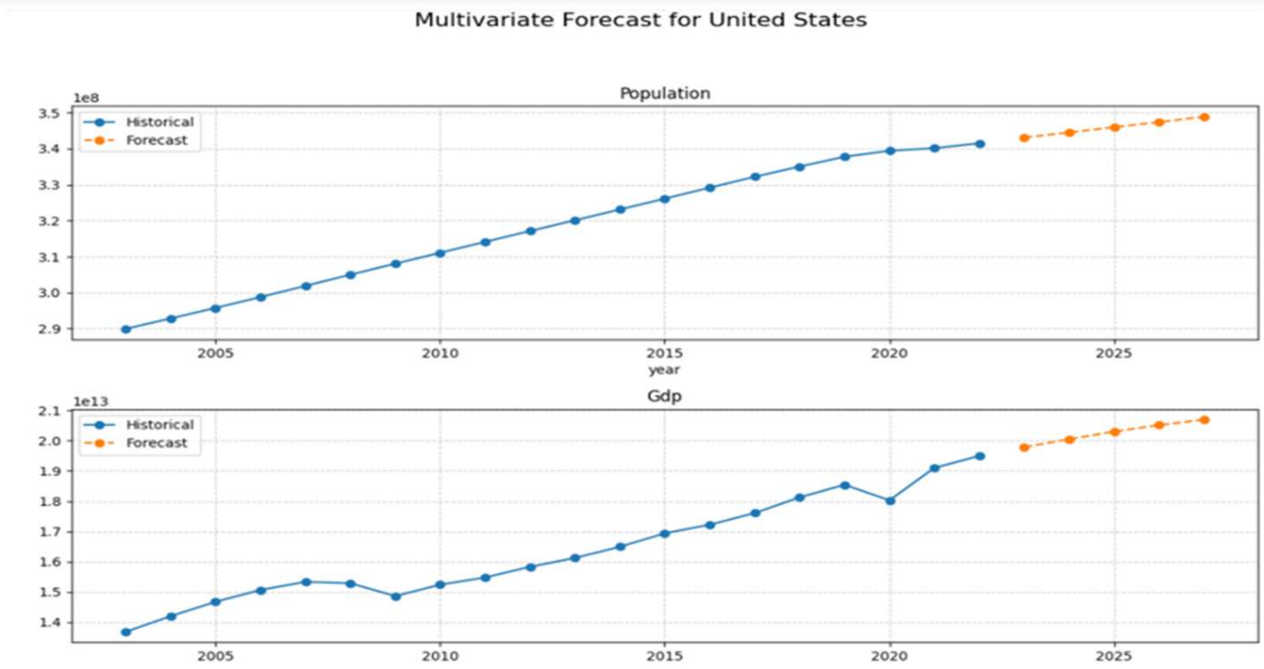
Prophet with Exogenous Variables

- The Key Insight: We must include the external drivers of emissions.
- We revisited the Prophet model, this time
  adding GDP and Population as external regressors.
- Recommendation: The Prophet model with tuned hyperparameters and multiple regressors (GDP, Population) has a significantly lower historical RMSE (46.7) compared to the model with no regressors (362.2).
- Inference: By including the projected growth of GDP and Population, the model forecasts a continued rise in GHG emissions, aligning much more closely with the observed historical trend.



Modeling Approach 4: Multivariate Time Series Models

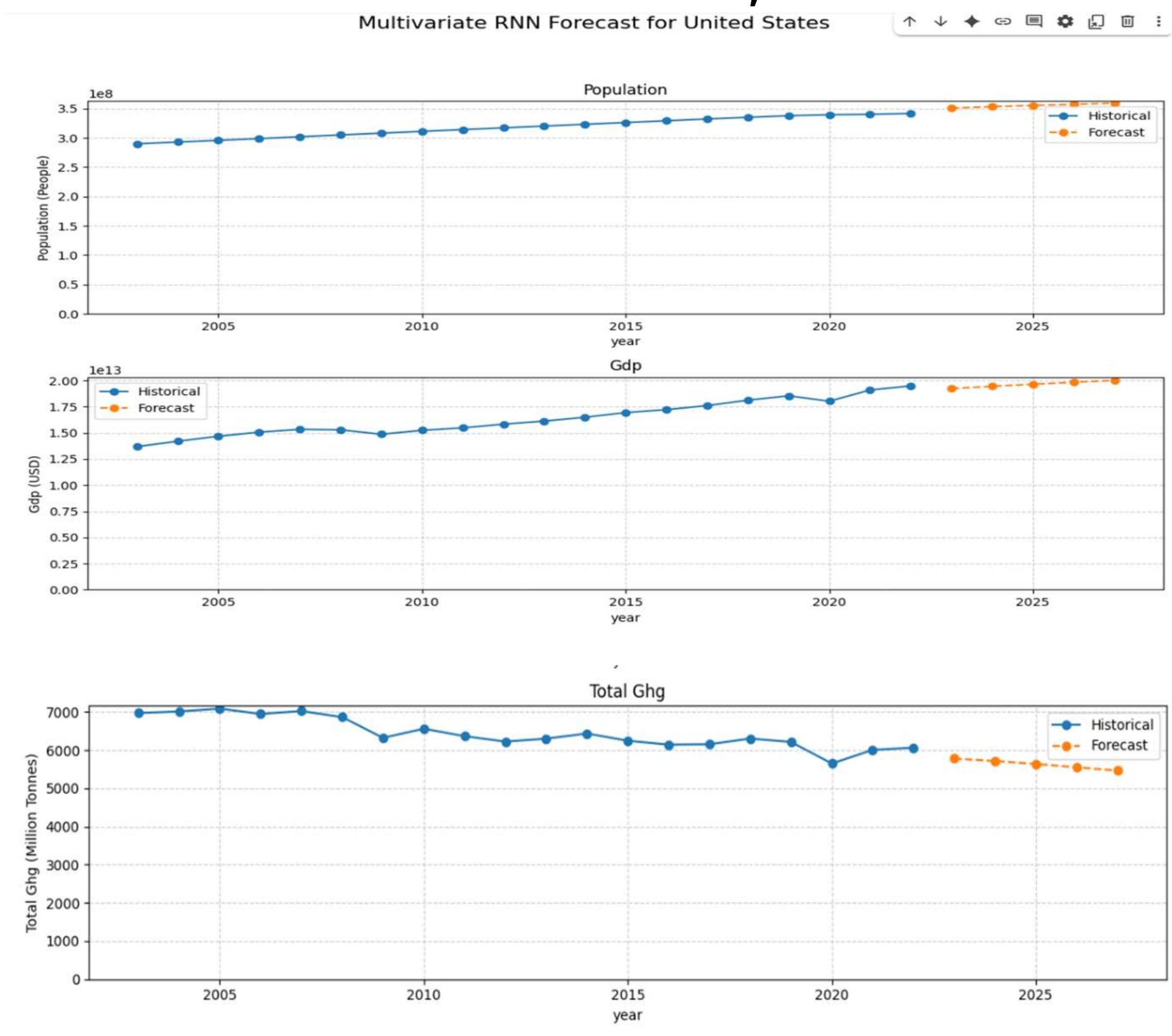
- These models can analyze multiple variables at once to understand their combined effect.
- VAR (Vector Autoregression): Models the linear interdependencies among multiple time series.
- LSTM (Long Short-Term Memory): A type of recurrent neural network (RNN) excellent at learning complex, non-linear patterns and long-term dependencies.
- The VAR model captures the two-way influence between Population, GDP, and Total GHG.
- Forecast: The model predicts a continued increase in Population and GDP, with GHG emissions fluctuating but remaining relatively stable. This reflects the complex interplay between the variables.



Image_D 22 : VAR forecast

Modeling Approach 4: Multivariate Time Series Models , continued…

- Results: LSTM Model
- The LSTM model learns the non-linear relationships from historical sequences of Population, GDP, and GHG.
- Forecast: The forecast shows a continued upward trend for all variables, extending the historical patterns based on the complex, non-linear dynamics learned by the model.



Model Performance Comparison

Model	Type	Key Feature	RMSE	R-squared	Forecast Behavior
Linear Regression	General ML	Linear	0.086	-1.245	Poor fit, unrealistic
Random Forest	General ML	Non-linear Ensemble	**0.056**	**0.040**	Good fit, realistic
ARIMA (1,1,0)	Univariate	Handles Trend	689.15	N/A	Flat forecast
ARMA (7,7)	Univariate	AR + MA terms	**533.11**	N/A	Better fit, but volatile
Prophet (No Exog)	Univariate	Decomposable	362.23	N/A	Plateau forecast
Prophet (Exog)	Multivariate	Regressors	**46.72**	N/A	**Follows Trend**
VAR	Multivariate	Linear System	N/A	N/A	Dynamic forecast
LSTM	Multivariate	Non-linear Sequence	N/A	N/A	Follows Non-linear Trend

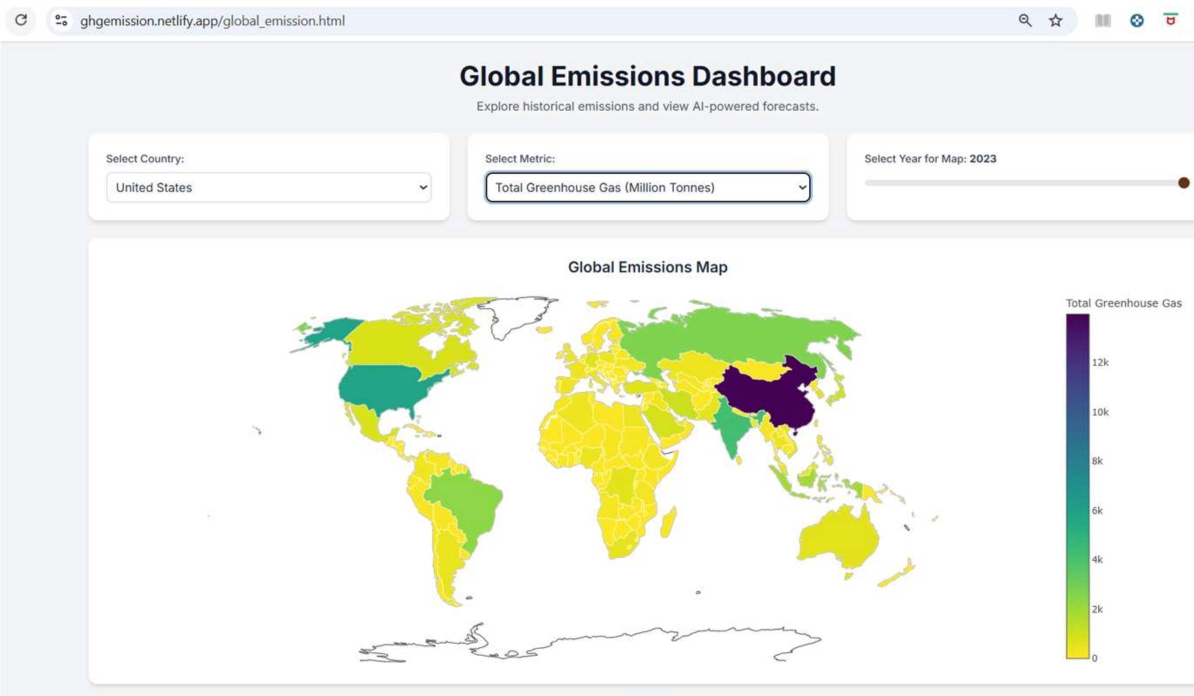
Univariate models (ARIMA, Prophet without regressors) are insufficient. They fail to predict future trends because the year-over-year changes in emissions are too random (like white noise).

Multivariate models (VAR, LSTM, Prophet with regressors) are superior. They generate more plausible forecasts by incorporating the external factors known to drive emissions.

The Prophet model with exogenous variables provided the best fit to the historical data, highlighting the importance of including drivers like GDP and population for accurate forecasting.

Dashboard 1: Global Emissions Dashboard

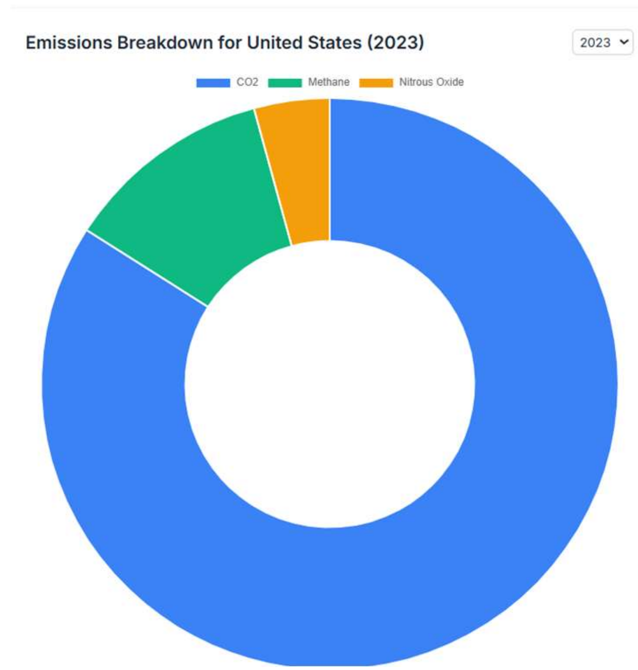
https://ghgemission.netlify.app/global_emission.html



historical emissions and view AI-powered forecasts for any country.

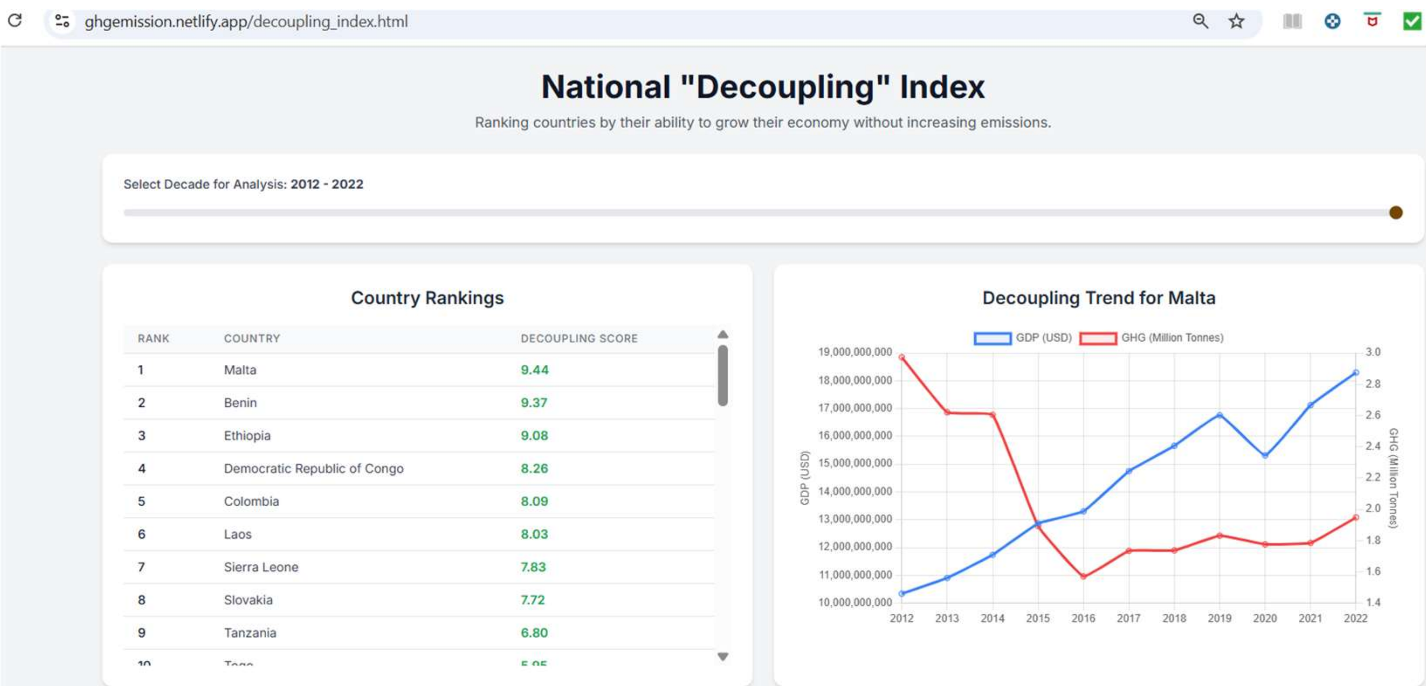
Visualize global emissions on an interactive map.

Analyze emissions breakdowns by gas (CO_2, Methane, Nitrous Oxide).



Dashboard 2: National "Decoupling" Index

https://ghgemission.netlify.app/decoupling_index.html



Purpose: Ranks countries by their ability to grow their economy without increasing emissions.

Decoupling Score = (Avg Annual GDP Growth %) - (Avg Annual GHG Growth %)

A high positive score indicates successful decoupling.

Dashboard 3: "What-If" Analysis

https://ghgemission.netlify.app/what_if

A simulation tool to see how changing assumptions about future GDP and population growth affects GHG emission predictions.

Users can adjust sliders to create different hypothetical scenarios.

Powered by the trained multivariate LSTM model.

Conclusion

In essence, models that incorporate the external factors known to drive CO2 emissions (such as GDP, population, and energy use) perform better and generate more plausible forecasts of future emission trajectories than models that rely solely on the historical pattern of the CO2 series itself.

Future Directions

•Integrate real-time and higher-frequency data (e.g., daily satellite imagery).

•Expand the model to include more granular data (e.g., specific crop types, soil data).

•Incorporate the impact of climate policies and technological advancements as model features.

•Develop more sophisticated non-linear models to capture even more complex interactions.

Impact: How This Analysis Can Be Used

Policy Making: Assess the effectiveness of climate policies and set future reduction targets.

Climate Modeling: Help scientists build and refine models to predict climate change scenarios.

Industry Planning: Allow businesses in agriculture to anticipate regulations and adapt practices.

Public Awareness: Visualize trends to highlight the severity and sources of climate change.

Investment & Development: Guide investment in sustainable agriculture and emission reduction projects.